

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q12-0157

Rockdale 138 kV
May 04, 2014

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q12-0157. The Interconnection Customer proposes a 248.8 MW natural gas combined cycle facility interconnecting at the Rockdale 138 kV substation in the Pennsylvania area. The request was received on September 13, 2012. The study evaluated system intact and single contingency conditions using the regional planning case, including 110 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 10 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$93,306,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of FT HUMBUG to ORCHARD RD 500.0 kV, the FORT HUMBUG 500-345 kV Autotransformer loads to 131.5% of its 628 MVA rating. Project DFAX: 5.6%.

For the loss of MCKENZIE RIVER to ROCKDALE 138.0 kV L, the ROBERTS CRK SUBSTATION to ROCKDALE 138.0 kV loads to 129.1% of its 188 MVA rating. Project DFAX: 100.0%.

For the loss of ROBETS CREEK to ROCKDALE 138.0 kV, the MCKENZIE RIVER to ROCKDALE 138.0 kV L loads to 126.8% of its 191 MVA rating. Project DFAX: 100.0%.

For the loss of HILLTOP to TALENENERGY MARTINS CRK LLC WILLIAMSPORT 500.0 kV, the HILLTOP 500 kV/345 kV TX loads to 125.4% of its 806 MVA rating. Project DFAX: 8.7%.

For the loss of HILLTOP 500-345 kV Autotransformer, the TALENENERGY MARTINS CREEK LLC WILLIAMSPORT 500-345 kV Autotransformer loads to 122.9% of its 821 MVA rating. Project DFAX: 7.2%.

For the loss of FORT HUMBUG to ORCHARD RD 500.0 kV, the FT HUMBUG to W CAMPUS 345.0 kV Line loads to 121.6% of its 1,371 MVA rating. Project DFAX: 10.0%.

The LITTLE CHUTE to TALENENERGY MARTINS CREEK LLC HARRISBURG 345.0 kV is overloaded to 120.3% of its 1,012 MVA emergency rating for the loss of the CORONET to FORT HUMBUG 500.0 kV. The project contributes 8.3% DFAX (20.8 MW) to this overload.

PINEY CRK PROJECT 500-345 kV Auto: 119.5% loading (776 MVA) following the outage of the HILLTOP to PINEY CRK PROJECT 500.0 kV Line; DFAX = 5.9%.

The AMERICAN to URBAN 345.0 kV is overloaded to 113.1% of its 1,270 MVA emergency rating for the loss of the FT HUMBUG to ORCHARD RD 500.0 kV. The project contributes 5.0% DFAX (12.5 MW) to this overload.

ROBERTS CREEK to CORNING 138.0 kV: 102.5% loading (227 MVA) following the outage of the MCKENZIE RIV to ROCKDALE 138.0 kV L; DFAX = 100.0%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
FORT HUMBUG 500-345 kV Autotransformer	Replace transformer with higher-rated unit	\$29,527,000	36% (\$10,475,000)
ROBERTS CRK SUBSTATION to ROCKDALE 138.0 kV	Rebuild line with higher-capacity conductor	\$1,605,000	100% (\$1,605,000)
MCKENZIE RIVER to ROCKDALE 138.0 kV line	Rebuild line with higher-capacity conductor	\$5,605,000	100% (\$5,605,000)
HILLTOP 500 kV/345 kV TX	Replace transformer with higher-rated unit	\$27,833,000	43% (\$11,969,000)
TALENENERGY MARTINS CREEK LLC WILLIAMSPORT 500 kV/345 kV TX	Replace transformer with higher-rated unit	\$23,763,000	41% (\$9,779,000)
FT HUMBUG to W CAMPUS 345.0 kV Line	Rebuild line with higher-capacity conductor	\$55,842,000	23% (\$13,052,000)
LITTLE CHUTE to TALENENERGY MARTINS CREEK LLC HARRISBURG 345.0 kV	Rebuild line with higher-capacity conductor	\$34,855,000	24% (\$8,519,000)
PINEY CRK PROJECT 500-345 kV Auto	Replace transformer with higher-rated unit	\$23,583,000	17% (\$3,988,000)
URBAN to AMERICAN SUBSTATION 345.0 kV line	Rebuild line with higher-capacity conductor	\$51,493,000	48% (\$24,590,000)
ROBERTS CREEK to CORNING 138.0 kV Line	Reconductor line	\$3,724,000	100% (\$3,724,000)
	Total allocated to project		\$93,306,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.